
Study guide for Midterm 1

Below I bring up various things we have studied. Many of these concepts are to be "known" by students in Stat 143 and Stat 243. Here, in Stat 343, I have emphasized the mathematical justifications for a number of these concepts, and left to the reading and exercises (some bracketed) others. The burden of being able to explain why things are true—arguing from definitions when necessary—should be taken on, whereas in a lower statistics course one might only feel the need to memorize facts.

Probability

- the terminology
 - sample space, events, set theory, complement, mutual exclusivity, etc.
 - random variable (discrete and continuous), pmf, pdf, cdf, mgf
- counting principles, binomial coefficients, etc. (see Table 2.3, p. 49)
- simulations and long-term relative frequency
- axioms of probability, and rules that follow from them; conditional probability, independence, law of total probability, Bayes' rule

Random variables

- computations of expected value, variance, other moments
- various families
 - discrete: Bernoulli, binomial, negative binomial, Poisson, geometric, hypergeometric, multinomial
 - continuous: uniform, exponential, normal, gamma, beta
 - the parameters that distinguishes one member of a family from another
 - scenarios in which one turns to a certain family
- joint distributions, independence, mean/variance when variables are combined, covariance, marginal and conditional distributions
- the cdf method (its purpose, and where the technique is useful)
It is used in Examples 3.1.7 through 3.1.10, in Exercise 3.10, and again in the proof of (4) in Lemma 3.4.6

Hypothesis testing

terms: null hypothesis, alternative hypothesis (one- and two-sided), test statistic, p -value, significance level, Type I/II errors, power of a test, rejection and non-rejection regions

Specific results to know and to prove(?)

- Lemma 2.5.4, p. 78
- Theorem 2.5.8, p. 80
- Lemma 2.5.9, p. 81 (See Exercise 2.67)
- Lemma 2.6.12, p. 89
- Lemma 2.6.14, p. 90, and the surprising(?) conclusion that follows from Exercise 2.88
- Exercises 2.9, 2.14, 2.19, and 2.20
- Example 3.1.13, p. 143
- Theorem 3.2.3, p. 146
- Lemmas 3.2.4 and 3.2.5, p. 146
- Theorem 3.3.6, p. 155

R commands

- know so you can write syntactically correct commands that apply to a given scenario:
`dbinom()`, `pbinom()`, `qbinom()`, `pnbinom()`, etc. (all the pmf/pdf, cdf and inverse cdf functions that exist for the families of r.v.s above)
- know so you can recognize what a given command is meant to accomplish and understand its output:
`binom.test()`, `gf_dist()`, `rbinom()`, `rexp()`, etc.